Verhoeff's Iron Hematoxylin Staining

Reagents:

1. Verhoeff's iodine (50 ml stock solution)

Iodine	- 1 g
Potassium iodide	- 2 g
Distilled water	- 50 ml

2. Verhoeff's iron hematoxylin (36 ml)

Hematoxylin (5% wt/vol) in absolute alcohol	- 20 ml
Ferric chloride (10% wt/vol)	- 8 ml
Verhoeff's iodine	- 8 ml

Mix a fresh solution, in the order given for each usage.

3. Ferric chloride (2% wt/vol)

4. Alcohol (95% vol/vol)

Procedure:

Step	Reagent / Procedure	Time
1	Acetone	5 minutes
2	Water	1 wash
3	Verhoeff's iron hematoxylin	20 minutes
4	Water	3 washes
5	Ferric chloride (2% wt/vol)	1 - 3 minutes
6	Water	3 wash
7	Microscopy	* If over-differentiated, go back to the step 3.
8	Ethanol (95% vol/vol)	30 seconds
9	Ethanol (100% vol/vol)	30 seconds
10	Xylene	3 minutes
11	Dry	
12	Permount and coverslip	

* Halt staining when nuclei and elastin fibers are still black with weak background.

NOTE: Chamber system (**Fig 1**) can be used for staining instead of Microprobe system.


Fig 1. Chamber system

Results

Elastic fibers - black
Nuclei - black

Reagents

Reagent	Source	Catalog #
Acetone	Fisher	S25120A
Iodine	Fisher	I35-100
Potassium Iodide	Sigma	P8256
Hematoxylin	Sigma	H9627
Ferric (III) Chloride	Sigma	F2877
Ethanol	Fisher	BP2818-4
Xylene	Millipore	XX0060-4

Reference

Cellular Pathology Technique, Fourth Edition. Culling, Allison, and Barr. Butterworths, Boston. 1985. p.168 - 75.

Protocol Developed : Daugherty Laboratory
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